

A Quantitative Model of Bank Merger Dynamics

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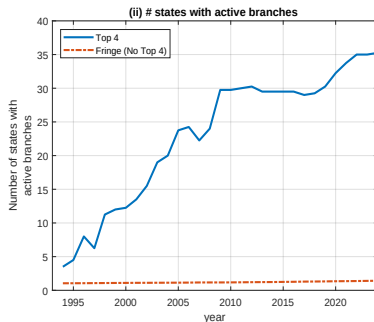
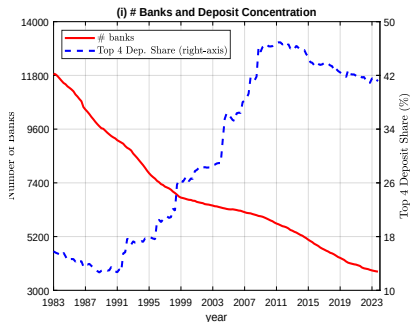
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* The views expressed in this paper do not necessarily reflect those of the Federal Reserve Bank of Philadelphia or the Federal Reserve System.

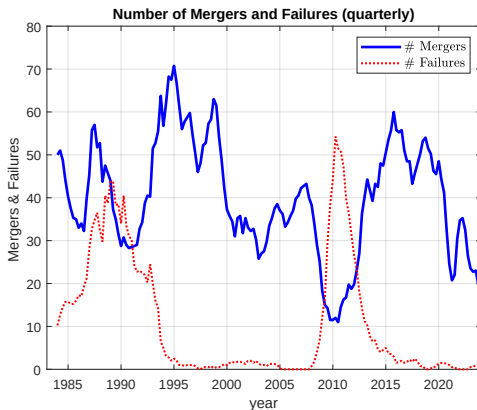
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INTRODUCTION



- ▶ Large decline in the number of banks and rise in banking concentration in the past 40 years due to a bank merger wave. [▶ Distn](#)
- ▶ Steady state market shares Pre-Riegle-Neal (1994) and post-crisis (2011).

BANK MERGERS DRIVE BANK CONSOLIDATION



- ▶ Largest number of mergers following Riegle-Neal (some states relaxed branching prior and there was significant numbers of mergers among regional banks following Dodd-Frank)

OUR QUESTION

“We document these facts because they are an important step in developing a theory of bank size distribution. Although we do not provide one, such a theory would be valuable because it could be used to answer important questions such as: How costly were the pre-1980 limits on bank size? How will the bank size distribution continue to evolve? And will there be more concentration? If so, should policy do anything about it?”

Janicki and Prescott (2006)

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- ▶ Our paper can be thought of providing one such theory.
- ▶ We use our model as a laboratory to quantitatively assess how merger regulation affects bank lending, markups, financial stability, and allocative efficiency?

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 - ▶ Include technology differences (e.g. IRS as in Diamond (1984)), entry/exit, ...

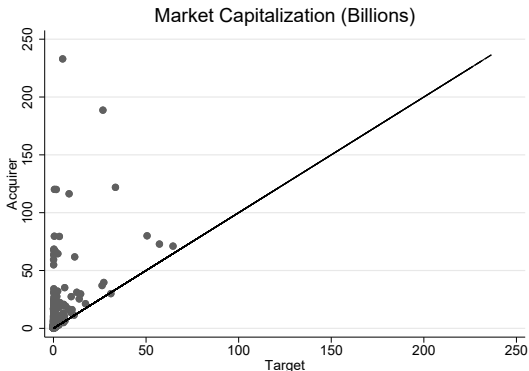
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3. After estimating parameters, we use the model as a laboratory to assess the implications of regulatory policy for:
 - ▶ the marginal propensity to lend across bank size
 - ▶ granularity of big bank lending (i.e. how much do idiosyncratic shocks to the top 4 contribute to aggregate bank lending)
 - ▶ allocative efficiency and financial stability
 - ▶ monetary policy effectiveness
 - ▶ nonbank competition

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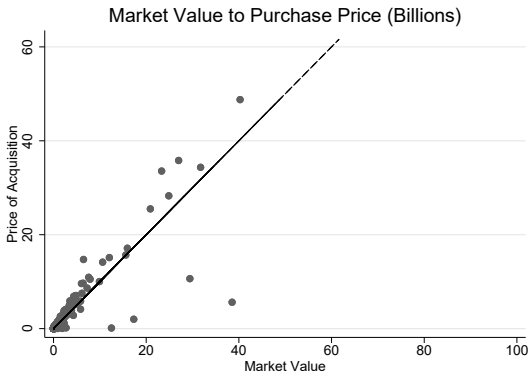
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4. We use the model to evaluate bank merger regulation:
 - ▶ Policy needs to account for possible benefits to allocative efficiency and financial stability tradeoff with costly markups

DATA: ACQUIRERS VS. TARGETS (PUBLIC BANKS)



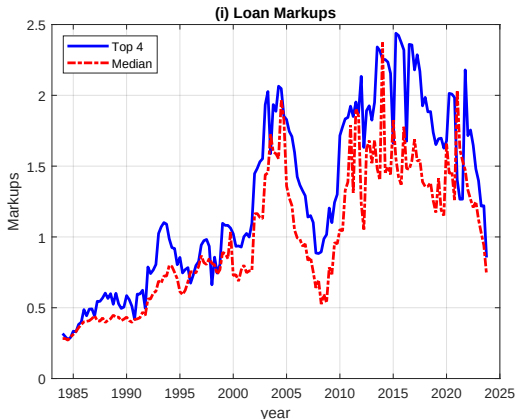
- ▶ Acquirers are larger than targets, whether measured by assets or market capitalization (1990 - 2019). [▶ By Assets](#)
- ▶ For public banks, the median acquirer is seven times larger than median target.

DATA: PRICE TO TARGET MARKET VALUE



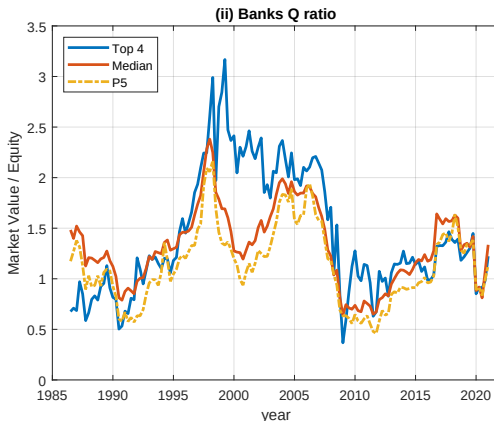
- ▶ We find that (target) size weighted mean of the ratio of merger price to target value is 1.11 and the median is 1.35.
- ▶ This suggests that larger banks have substantial bargaining power in the merger process.

DATA: BANK LOAN MARKUPS



- Rise in top 4 bank share and declining number of fringe banks coincides with a rise in markups.

DATA: TOBIN'S Q



Note: banks' Q is measured as market value over book equity as in Begenau et al. (2025)

- Rise in top 4 bank share and declining number of fringe banks coincides with a rise in the value (market to book value of equity) of the banking sector.

DATA: MARGINAL PROPENSITY TO LEND

$$\Delta L_{it} = \beta_{k(i)} \xi_{k(i)} X_{it} + \beta_{k(i)} \nu_{it} + \zeta_{it}.$$

	Dependent Variable $\Delta \log(L_{it})$			
	Full Period 1984-2019	Pre-reform 1984-1993	Transition 1994-2007	Post-GFC 2011-2019
ν_{it}	0.510*** (0.00101)	0.520*** (0.00170)	0.499*** (0.00165)	0.456*** (0.00256)
$\nu_{it} \times I_{t4}$	0.449*** (0.0359)	0.377*** (0.0698)	0.508*** (0.0490)	-0.237 (0.234)
N	946299	353150	355389	169536
R-sq	0.362	0.398	0.416	0.287

: Note: all regressions include Bank FE, Time FE (quarter-year), and other bank level controls.

- ▶ Results suggest the average MPL out of transitory shocks ν_{it} to deposits is 0.51, with the largest banks having a higher MPL of 0.96. ▶ MPL
- ▶ Dominant MPL declines over time contributing to granularity changes.

DATA: GRANULARITY AND AGGREGATE LENDING

- Applying Gabaix (2011) to banking, let the granular residual be:

$$\Gamma_t^d = \sum_{i=1}^K \omega_{it} \nu_{it} \quad (1)$$

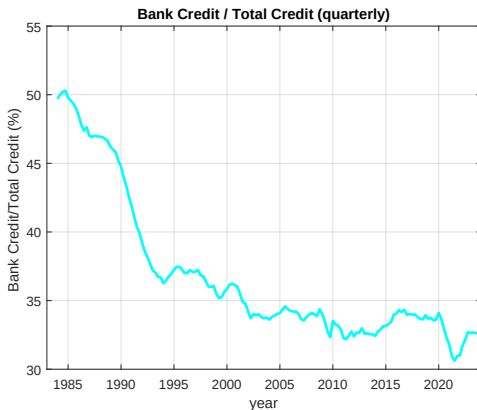
where ν_{it} is an idiosyncratic transitory deposit shock to the top $K = 4$ banks

- We compute the R^2 from the regression $\Delta L_t = \beta_{\Gamma} \Gamma_t^d + \epsilon_t^{\Gamma}$

	Top 4 Banks - Dep. Var. $\Delta \log(L_t)$			
	Full Period 1984-2019	Pre-reform 1984-1993	Transition 1994-2007	Post-GFC 2011-2019
N	144	40	56	36
R^2	0.100	0.104	0.212	0.084

- R^2 of 0.10 suggests that idiosyncratic shocks to the largest 4 banks explain 10% of the variation in aggregate bank lending. [► Gabaix](#)

BANK CREDIT TO TOTAL CREDIT



- ▶ A significant and growing presence of nonbanks in lending markets

DATA SUMMARY

1. Increase in deposit market concentration and decline in the number of banks post-Riegle-Neal is driven by mergers.
2. Increase in bank loans' markups over time and heightened Tobin's Q following deregulation.
3. Acquirers are much larger than targets and have substantial bargaining power.
4. Marginal propensity to lend (MPL) higher for top 4 until post-crisis
5. Granular Banks: deposit shocks to the largest 4 banks have a substantial effect on bank lending (and total credit, output).
6. A significant and growing presence of nonbanks in lending markets (share of bank lending in total credit drops from 50% to 33%).

MAPPING A SIMPLE MODEL TO THE DATA

- ▶ Rising market share of large banks along with declining numbers & large size difference between targets and acquirers → our model includes a merger stage building on Gowrisankaran and Holmes (2004).
 - ▶ a granular dominant bank acquires a *measure* of atomless (zero measure) fringe banks.
 - ▶ model nests standard heterogeneous firm dynamic framework where there are no spillovers between individual banks
- ▶ Granular Banks & Rising markups → model of Cournot competition in the loan market provides implications for the bank lending channel of monetary policy.
- ▶ Rising loan market competition from nonbanks → model adds non-bank sector with discrete choice problem as in Buchak et al. (2018).
- ▶ Financial (in)stability → model adds entry/exit creating an endogenous size distribution of banks.

MODEL ESSENTIALS

- ▶ **Banks and nonbanks** intermediate between large numbers of
 - ▶ **households** can deposit at a bank with deposit insurance, hold bank/nonbank stocks, and inject equity to banks and nonbanks ▶ deposit supply .
 - ▶ **borrowers** with a large risky project choose whether to take a loan from a bank $k = B$ or non-bank $k = N$ with costly project monitoring ▶ loan demand

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- ▶ **Merger stage**: dominant (d) bank makes TOLI offer to a measure of fringe (f) banks.
- ▶ **Loan stage**: dominant bank Stackelberg competes with fringe banks in the loan market, given nonbank interest rate r_N^L , to determine r_B^L . ▶ nonbanks
 - ▶ Dominant banks internalize their effect on their own loan return but impose a pecuniary externality on others.
 - ▶ Deposit insurance creates a moral hazard problem for banks' portfolio choice between risky loans and safe securities ($r_B^L > r^A > r^D$).

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- ▶ **Uncertainty**: Idiosyncratic shocks ϵ_i to deposits and charge-offs λ_i' . Aggregate shocks θ to the fraction of non-performing loans follow Markov process. ▶ SMM estimation ▶ Transition ▶ market shares

TIMING

The timing in any period is as follows:

- M1. Merger Stage: Starting in state θ in a market with measure γ of fringe banks, the dominant bank makes a take-it-or-leave-it merger bid to a measure $(\gamma - \Gamma)$ of fringe banks subject to regulatory costs of doing so in order to grow its deposit base $D_d = D_d^- + (\gamma - \Gamma)D_f$.
- M2. Loan Stage: Banks receive temporary shock ϵ_i to deposits. Banks allocate their deposits $D_i + \epsilon_i$ to the loan market L_i or securities $D_i + \epsilon_i - L_i$. Dominant banks move first in the loan market followed by the fringe banks and nonbanks.
- M3. Payoff Stage: At the end of the period, loan defaults θ' and charge-offs λ'_i are realized. The (eop) profit of bank i type $j(i) \in \{d, f\}$ is given by:

$$\begin{aligned} \pi_i(\mathbf{L}_i; \epsilon_i, S, \theta', \lambda'_i) = & [\theta' r_B^L - (1 - \theta') \lambda'_i] \mathbf{L}_i + r^A (D_i + \epsilon_i - \mathbf{L}_i) \quad (2) \\ & - r^D (D_i + \epsilon_i) - C_i(\mathbf{L}_i) - \kappa_i \end{aligned}$$

Exit and entry decisions are made and dividends paid.

MERGER STAGE

- ▶ Starting in state $s = (\gamma, D_d, \theta)$ where γ is bop measure of fringe banks and D_j are bop deposits of $j \in \{d, f\}$ type banks.
- ▶ The dominant bank makes a TOLI offer of price $p(\Gamma, s)$ per unit of deposits to a measure $(\gamma - \Gamma)$ of fringe banks where Γ is their measure following the merger state.
- ▶ The dominant bank solves the following problem

$$v_d(s) = \max_{\Gamma \in [0, \gamma]} p(\Gamma, s) D_f - H_n(\Gamma, s) + \int w_d(S, \epsilon_d) dF_d(\epsilon) \quad (3)$$

subject to

$$p(\Gamma, s) D_f \geq \int w_f(S, \epsilon_f) dF_f(\epsilon) \quad (4)$$

where $S = (\Gamma, D'_d, \theta)$, $D'_d = D_d + (\gamma - \Gamma) D_f$, $w_f(S, \epsilon_f)$ is the value of a target fringe bank who chooses not to accept the merger offer, and $H_n(\Gamma, s)$ are regulatory constraints on merger choice Γ . ▶ RegConstraint

red=endogenous variable, blue=model parameter

LOAN STAGE

Taking Merger Stage endogenous state S as given in this stage, the Stackelberg problem of the dominant bank is

$$w_d(S, \epsilon_d; r_N^L) = \max_{L_d \leq D'_d + \epsilon_d} \mathbb{E}_{\theta'|\theta} \left[\max_{x'_d \in \{0,1\}} (1 - x'_d) (\mathcal{D}_d(L_d; \epsilon_d, S, \theta', \lambda'_d) + \beta v_d(s')) \right], \quad (5)$$

st

$$\mathcal{L}_B^S \equiv L_d + \Gamma \cdot L_f^*(L_d, S, r_N^L) = \mathcal{L}_B^D(\theta, r_B^L, r_N^L), \quad (6)$$

$$\gamma' = F(L_d, S, \theta'). \quad (7)$$

- ▶ Eqn (6) illustrates both loan market clearing and that the dominant bank internalizes how their loan choice affects fringe loan decision rules, $L_f^*(\cdot)$, and the bank loan interest rate, r_B^L .
- ▶ The law of motion for the distribution of fringe banks in eqn (7) ensures that the dominant bank internalizes its impact on future market structure.
- ▶ Endogenous evolution of banking industry state:

$$s = (\gamma, D_d, \theta) \rightarrow S = (\Gamma, D'_d, \theta) \rightarrow s' = (\Gamma, D'_d, \theta')$$

red=endogenous variable, blue=model parameter

MARKOV PERFECT EQUILIBRIUM

- For given r^A and r^D , a **Markov Perfect Cournot Equilibrium** is a set of bank value functions $\{v_i, w_i\}$ and policy functions $\{\Gamma, L_i, \ell_f, x'_i, \chi'_f, e'_i\}$ for $i \in \{d, f\}$, ℓ_N , household value V_H , and policy functions $\{A_H, D_H, \{\Phi_i\}_{i=d,f}, \Phi_N\}$, borrower value Π_E and choices $\{\iota, k\}$, prices $\{p, r_B^L, r_N^L\}$, stock prices $\{P_i\}_{i=d,f}$, and P_N , tax function τ , and transition functions for $\{\gamma', D'_d\}$ such that:
1. The pre-merger value function v_d solves (3). The merger quantity Γ maximizes (3).
 2. The merger pricing function p satisfies (4).
 3. Given $\{r_B^L, r_N^L\}$, $\{\iota, k\}$ are consistent with entrepreneur optimization
 4. Given $\Upsilon_D = (r_A - r_D)$, $\{P_i\}_{i=d,f}$, and P_N , $\{A_H, D_H, \{\Phi_i\}_{i=d,f}, \Phi_N\}$ are consistent with hh optimization
 5. The post-merger value functions w_d and w_f solve (5) and (17). Bank policy functions $(\{L_d, \ell_f, x'_d, \chi'_f\})$ are consistent with bank optimization ((5))-((17))
 6. Consistency requires $\ell_f = L_f$ and $\chi'_f = x'_f$.
 7. r_B^L clears the loan market (6).
 8. r_N^L satisfies the nonbank first order condition (20).
 9. The mass of entrants γ^e solves the entry problem (18)
 10. Transition functions are consistent with mergers, entry, and exit (17)
 11. Taxes and transfers satisfy the government budget constraint

MARGINAL PROPENSITY TO LEND

- The first order condition for L_i trades off marginal benefits and costs given by:

$$\begin{aligned} \mu_i(\epsilon_i, S) &= E_{\theta', \lambda'_i \leq \lambda_i^{x'_i=1} | \theta} \left\{ [\theta' r_B^L - (1 - \theta') \lambda'_i] + \left[\theta' \frac{\partial r_B^L}{\partial \mathcal{L}_B} \frac{\partial \mathcal{L}_B}{\partial L_i} L_i \right] + \left[\beta \frac{\partial v_i(s')}{\partial s'} \frac{\partial s'}{\partial L_i} \right] \right\} \\ &- E_{\theta', \lambda'_i \leq \lambda_i^{x'_i=1} | \theta} \left\{ \left[\frac{\partial \Pr(\pi_i < 0, x'_i = 0)}{\partial L_i} \frac{\partial \psi_i(|\pi_i|)}{\partial |\pi_i|} \right] + r^A + \frac{\partial C_i(L_i)}{\partial L_i} \right\} \end{aligned}$$

where μ_i is the multiplier on the insured financing constraint that $L_i + A_i \leq D_i + \epsilon_i$ and $\lambda_i^{x'_i=1}$ is bank i 's exit threshold for chargeoffs.

- μ_i determines whether MPL is either 0 or 1 across different states.
- In some states (e.g. $\epsilon_i < 0$), banks behave like “hand-to-mouth” consumers.

► ex-post profit

► L_f vs ψ_f

MARGINAL PROPENSITY TO LEND

$$\begin{aligned} \mu_i &= E_{\theta', \lambda'_i \leq \lambda_i^{x'=1} | \theta} \left\{ [\theta' r_B^L - (1 - \theta') \lambda'_i] + \left[\theta' \frac{\partial r_B^L}{\partial \mathcal{L}_B} \frac{\partial \mathcal{L}_B}{\partial L_i} L_i \right] + \left[\beta \frac{\partial v_i(s')}{\partial s'} \frac{\partial s'}{\partial L_i} \right] \right\} \\ &- E_{\theta', \lambda'_i \leq \lambda_i^{x'=1} | \theta} \left\{ \left[\frac{\partial Pr(\pi_i < 0, x'_i = 0)}{\partial L_i} \frac{\partial \psi_i(|\pi_i|)}{\partial |\pi_i|} \right] + r^A + \frac{\partial C_i(L_i)}{\partial L_i} \right\} \end{aligned}$$

→ Positive effect from expected return on loans

MARGINAL PROPENSITY TO LEND

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→ Dominant banks internalize the impact of their lending on interest rates (zero for fringe banks)

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- Dominant bank lending affects continuation values and the distribution of banks: lower returns affects bank failure (this effect is also zero for fringe banks)

MARGINAL PROPENSITY TO LEND

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→ Cost of funding negative cash flows via seasoned equity, as in corporate finance lit., more costly for fringe banks

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→ Opportunity cost of loans

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→ Marginal cost of screening/monitoring loans

MODEL MARGINAL PROPENSITY TO LEND

- ▶ We run MPL regressions in the model in pre and post Riegle-Neal environments.

Coefficient	Pre-Riegle-Neal	Post-Riegle-Neal	Post-Dodd-Frank
β	0.28 (0.046)	0.277 (0.025)	0.473 (0.029)
$\beta \times 1_D$	0.714 (0.07)	-0.234 (0.248)	-0.465 (0.049)
R^2	0.999	0.013	0.987
N	22500.0	30064.0	14810.0

- ▶ As dominant bank grows larger with mergers, the loan funding constraint is less likely to bind and its MPL declines (consistent with data)

GRANULAR REGRESSIONS - BANK LENDING

- ▶ We run granularity regressions in the model in pre and post Riegle-Neal environments.

Coefficient	Pre-Riegle-Neal	Post-Riegle-Neal	Post-Dodd-Frank
Intercept	-0.0	-0.005	-0.001
Shock Coefficient	1.469	0.09	0.091
R^2	0.522	0.022	0.12
N	9000.0	15000.0	9000.0

- ▶ Model explains 52% (Pre) and 12% (Post) of bank lending variation (> data) ▶ total cred. ▶ data
- ▶ While market share of dominant banks increase, the decline in R^2 is due to the decline in MPL

ALLOCATIVE EFFICIENCY

- ▶ We use the following decomposition of weighted average bank-level cost, as proposed by Olley and Pakes (1996).

$$\hat{c} \equiv \sum_{i \in \{D, F\}} C_i(L_i) \omega(L_i) = \bar{c} + Cov(C_i(L_i), \omega(L_i)) \quad (8)$$

where $\omega(L_i)$ is the loan market share of bank i .

Moment	Data		Model	
	Pre Riegle-Neal	Post Dodd-Frank	Pre Riegle-Neal	Post Dodd-Frank
Avg (loan-weighted) mg net exp. $\hat{c}$	0.0178	0.0119	0.0141	0.0121
Avg mg net exp $\bar{c}$	0.0244	0.0198	0.0156	0.0159
$Cov(C'_i(L_i), \omega_i)$	-0.0065	-0.0079	-0.0015	-0.0033
Total Bank Loans $(L_d + \Gamma L_f)$			0.8388	0.7632

- ▶ Allocative efficiency, as measured by $\downarrow Cov(c, w)$ (i.e. lower cost banks provide a larger share of loans), increases as dominant banks can exploit increasing returns to scale.

COUNTERFACTUAL: EFFECTS OF RIEGLE-NEAL

Moment	Pre RN	Post RN (short-run)	Post RN (long-run)
Deposit Share Top 4	13.61%	31.24%	40.65%
Loan Share Top 4	18.58%	28.31%	31.25%
Bank Lending Share	44.44%	43.67%	43.01%
Bank Interest Rate	4.83%	4.92%	4.97%
Loans/Deposits Top 4	99.98%	67.18%	53.42%
Loans/Deposits Fringe	70.13%	75.29%	81.24%
Fringe Failure Rate	0.49%	0.11%	0.12%
Merger Rate	0.00%	2.10%	0.00%
Loan Markup Top 4	4.27	4.32	4.43
Loan Markup Fringe	1.87	1.84	1.72
Tobin's Q Top 4	0.96	6.09	5.73
Tobin's Q Fringe	0.71	1.20	1.27

- ▶ Allowing mergers ("Post RN" short-run/long-run) leads to a more concentrated banking sector, higher interest rates, and Tobin's Q , and lower failure rates
- ▶ Markups increase for Top 4 but decline for fringe as they operate at a higher marginal cost (fringe loan/deposit ratio increases) ▶ Dodd-Frank decomposition
- ▶ Effects on r_B^L limited by rising share of nonbanks (regulatory arbitrage accounts for over 20% of rising nonbank share)

CONCLUSION

- ▶ After documenting evidence on bank mergers, granularity, and the marginal propensity to lend in the data, we build a dominant-fringe model of imperfect competition with endogenous mergers to study the effects of regulation on:
 - ▶ market power
 - ▶ allocative efficiency and financial stability
 - ▶ the effectiveness of monetary policy
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- ▶ Contractionary monetary policy has smaller effects on bank lending post-RN and can contribute to rising mergers

▶ Monetary Policy

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- ▶ More nonbank competition increases financial instability absent mergers since the possibility of getting acquired leads fringe banks to take on less risk.

▶ Nonbank Competition

CONCLUSION

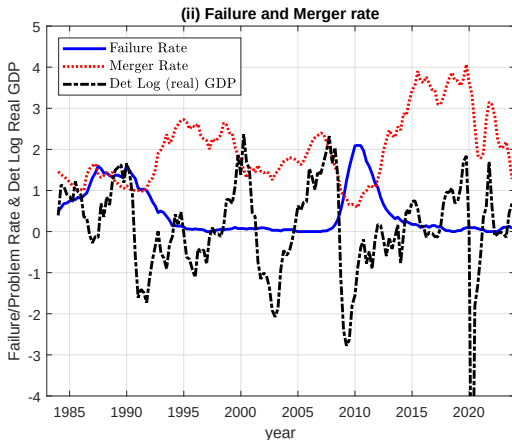
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- ▶ More nonbank competition increases financial instability absent mergers since the possibility of getting acquired leads fringe banks to take on less risk. ▶ Nonbank Competition
- ▶ We study the normative effects of Riegle-Neal (and alternative merger policies) taking into account the tradeoff between market power, financial stability, and allocative efficiency ▶ welfare



Appendix



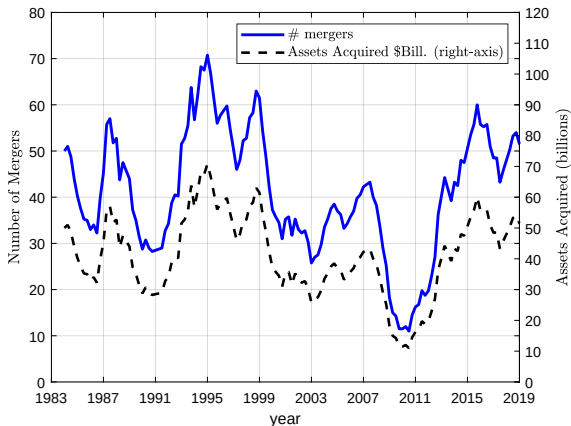
FAILURE AND MERGER RATE OVER BC



- ▶ Mergers are procyclical and Failures countercyclical
- ▶ Higher merger rate post- crisis than post-Riegle-Neal consistent with smaller denominator (number of banks)

[▶ Back](#)

NUMBER OF MERGERS AND ASSETS ACQUIRED



- Book value of Mergers measured using total assets follows similar pattern as the number of transactions

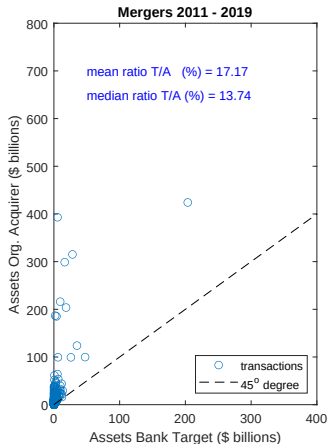
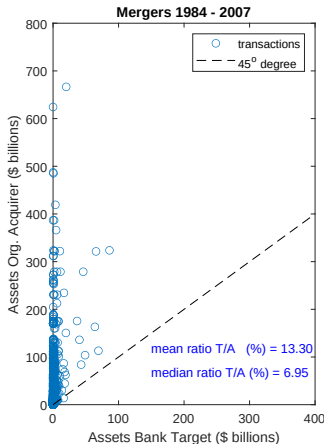


DISTRIBUTION ASSETS TARGET/ACQUIRER

Mergers 1984 - 2019	
Percentile	Ratio Target / Acquirer (%)
1%	0.07
5%	0.30
10%	0.67
25%	2.74
50%	9.25
75%	20.28
90%	34.33
95%	43.32
99%	63.68
Mean	14.48
Std Dev.	26.69



ACQUIRERS VS. TARGETS PRE / POST



- ▶ Period 1984 - 2007: for the median merger the acquirer is 14 times larger than the target
- ▶ Post GFC (2011-2019): for the median merger, the acquirer is 7 times larger than the target

[▶ return](#)
[▶ Back](#)



BANK MARGINAL PROPENSITY TO LEND

- We are interested in estimating the MPL $\beta_{j(i)}$ for bank i where $j(i) \in \{d, f\}$:

$$\Delta L_{it} = \beta_{j(i)} \Delta D_{it} + \zeta_{it} \quad (9)$$

- Following the MPC literature, the deposit shock process is given by:

$$\begin{aligned} \log(D_{it}) &= \xi_{j(i)} X_{it} + z_{it} + \varepsilon_{it} \quad \text{where} \quad z_{it} = z_{it-1} + \eta_{it} \\ \Rightarrow \Delta D_{it} &= \xi_{j(i)} \Delta X_{it} + \underbrace{\eta_{it} + \Delta \varepsilon_{it}}_{=\nu_{it}} \end{aligned} \quad (10)$$

- Then the deposit process in (10) implies that equation (9) can then be written as

$$\Delta L_{it} = \beta_{j(i)} \xi_{j(i)} \Delta X_{it} + \beta_{j(i)} \nu_{it} + \zeta_{it}. \quad (11)$$

so that the idiosyncratic shocks ν_{it} help identify the MPL.

► Return MPL



BANK GRANULARITY REGRESSIONS AND MPL

- $\Delta L_t = \beta_\Gamma \Gamma_t^d + \epsilon_t^\Gamma$ (Gabaix (2011)) can be rewritten as:

$$\Delta L_t \approx \sum_{i=1}^N \omega_{it} \Delta L_{it} = \beta_d \underbrace{\sum_{i=1}^K \omega_{it} \nu_{it}}_{\Gamma_t^d} + \beta_f \underbrace{\sum_{i=K+1}^N \omega_{it} \nu_{it} + \sum_{i=1}^N \omega_{it} (\beta_k \xi_k X_{it} + \zeta_{it})}_{=\epsilon_t^\Gamma}$$

- Under this approximation, $\beta_\Gamma \approx \beta_d$.
- Our estimate of β_Γ is a downward biased estimate of dominant bank MPL: estimated value from ΔL_t on Γ_t^d equals 0.474 vs MPL 0.889 (intuition derives from negative correlation of shocks, “deposit stealing argument”)

► Back



HOUSEHOLDS

- ▶ A mass $\mathcal{H}$ of identical, infinitely lived, risk neutral households with discount factor β are endowed with y units of the good each period.
- ▶ Households have access to securities that yield $1 + r_A$ with $r_A \geq 0$ and $\beta(1 + r_A) = 1$.
- ▶ They can also choose to supply their endowment to a bank, a nonbank, or to an individual borrower.

$$V_H(\{\theta, \Phi_i\}_{\forall i}, \Phi_N) = \max_{\{A_H, D_H, \{\Phi'_i\}_{\forall i}, \Phi'_N\}} E[C + \Upsilon_D D_H + \beta V_H(\{\Phi'_i\}_{\forall i}, \Phi'_N)]$$

subject to

$$A_H + D_H \leq y \tag{12}$$

$$C + A_H + D_H + \int [P_i + \mathbf{1}_{\{e_i=1\}} \cdot (K_i + \psi_i^e(K_i))] \Phi_i di + \Phi'_N P_N \tag{13}$$

$$= y + (1 + r_A)A_H + (1 + r_D)D_H + \int (\mathcal{D}_i + P_i) \Phi_i di + (\mathcal{D}_N + P_N) \Phi_N - \tau,$$



DEMAND FOR LOANS

- ▶ A mass $\mathcal{E}$ of ex-ante identical borrowers choose whether to fund a risky project subject to limited liability and if so, whether to do so through a bank B at rate r_B^L or a nonbank at rate r_N^L .
- ▶ The discrete choice of which lender to choose is subject to an unobservable extreme value shock and lender quality.
- ▶ As in the discrete choice literature, the bank/nonbank choice implies market shares for lender of type k :

$$\psi_k(\theta, r_B^L, r_N^L) = \frac{\exp(\alpha E_{\theta'|\theta} [\pi_E(r_k^L, \theta')] + \xi_k)}{\sum_{\hat{k} \in \{\mathcal{B}, \mathcal{N}\}} \exp(\alpha E_{\theta'|\theta} [\pi_E(r_{\hat{k}}^L, \theta')] + \xi_{\hat{k}})}.$$

where $\pi_E(r_k^L, \theta')$ is the borrowers project return conditional on borrowing.

- ▶ Bank loan demand

$$L_B(\theta, r_B^L, r_N^L) = \psi_B(\theta, r_B^L, r_N^L) L(\theta, r_B^L, r_N^L) \quad (14)$$

where $L(\theta, r_B^L, r_N^L)$ is the total demand for credit conditional on the aggregate state of the economy θ .



MERGER STAGE - OPTIMAL Γ

- ▶ Taking a first order condition with respect to Γ and assuming the participation constraint binds, we get:

$$\frac{\partial w_d(\Gamma)}{\partial \Gamma} = \frac{\partial w_f(\Gamma)}{\partial \Gamma}(\gamma - \Gamma) + \frac{\partial H(\Gamma)}{\partial \Gamma} - w_f(\Gamma, D'_d(\Gamma), \theta)$$

- ▶ The LHS is the marginal benefit for the dominant bank. It depends on how the acquisitions limits competition and improves the dominant bank profitability.
- ▶ The RHS is the marginal cost of the merger, which incorporates both the marginal cost due to greater merger regulation (H) and due to rising fringe value functions (w_f).

▶ return



LOAN STAGE: PROFITS AND DIVIDENDS

- ▶ The (eop) profit of bank i type $j(i) \in \{d, f\}$ is given by:

$$\pi_i(\mathbf{L}_i; \epsilon_i, \mathbf{S}, \theta', \lambda'_i) = [\theta' r_B^L - (1 - \theta') \lambda'_i] \mathbf{L}_i + r^A (D_i + \epsilon_i - \mathbf{L}_i) - r^D (D_i + \epsilon_i) - C_i(\mathbf{L}_i) - \kappa_i \quad (15)$$

where $C_i(L_i)$ is the cost of loan monitoring and κ_i are fixed costs of running the bank.

- ▶ Size dependent monitoring and operating costs are calibrated to be consistent with IRS as in data (e.g. Berger and Udell (1994) and Corbae and D'Erasmus (2025)) and theory (Diamond's (1983) delegated monitoring model).
- ▶ Denote dividends by:

$$\mathcal{D}_i(L_i; \epsilon_i, S, \theta', \lambda'_i) = \pi_i(L_i; \epsilon_i, S, \theta', \lambda'_i) - \mathbf{1}_{\{\pi_i(\cdot) < 0\}} \psi_i(|\pi_i(L_i; \epsilon_i, S, \theta', \lambda'_i)|) \quad (16)$$

where ψ_i is the cost of seasoned equity issuance.

- ▶ size dependent costs are calibrated along the lines of corporate finance theory.



FRINGE BANK LOAN PROBLEM

- The problem of the fringe bank is

$$w_f(L_d, S, \epsilon_f; r_N^L) = \max_{\ell_f \leq D_f + \epsilon_f} \mathbb{E}_{\theta', \lambda'_f | \theta} \left[\max_{\chi'_f \in \{0,1\}} (1 - \chi'_f) (\mathcal{D}_f(\ell_f; L_d, \epsilon_f, S, \theta', \lambda'_f) + \beta v_f(s')) \right]$$

- The total supply of loans by fringe banks is

$$\Gamma L_f^*(L_d, S, r_N^L) = \Gamma \int \ell_f(L_d, S, \epsilon_f; r_N^L) dF_d(\epsilon)$$

- The evolution of the mass of fringe banks is given by:

$$\gamma' = F(L_d, S, \theta') = \Gamma - \gamma^x(L_d, S, \theta') + \gamma^e(L_d, S, \theta') \quad (17)$$

- The evolution of dominant bank deposits is (assuming entrant takes over D_d)

$$D'_d = (1 - x'_d(L_d, S, r_N^L, \theta')) D_d + x'_d(L_d, S, r_N^L, \theta') (e'_d(L_d, S, r_N^L, \theta') D_d)$$

- The problem of the entrant is

$$v_i^e(L_d, S, r_N^L, \theta'; \gamma^e(x'_d, e'_d)) = \max_{e'_i \in \{0,1\}} (1 - e'_i) \{-K_i + v_i(s')\}, \quad (18)$$



NONBANK LOAN PROBLEM

- ▶ Identical nonbanks discount the future at rate $\beta = 1/(1 + r^D)$ and maximize discounted flow of dividends (no equity issuance costs)
- ▶ The profit of the representative nonbank is given by:

$$\pi_N(\ell_N, S, \theta', \lambda_N) = [\theta' r_N^L - (1 - \theta')\lambda_N - c_N]\ell_N \quad (19)$$

- ▶ The first order condition of the non-bank with respect to ℓ_N is given by

$$r^D = E_{\theta'|\theta} [\theta' r_N^L - (1 - \theta')\lambda_N] - c_N. \quad (20)$$

- ▶ Linearity is crucial for simplicity $\rightarrow$ perfectly elastic supply by nonbanks implies $r_N^L(\theta)$ consistent with countercyclical real interest rates conditional on a given r^A and r^D .



MODEL ESTIMATION - SMM

- ▶ We estimate most model parameters using SMM to match the banking industry data pre-Riegle-Neal (1984-1993). [▶ Parameters & Targets](#)
- ▶ Broadly we target moments related to [▶ Moments](#)
 - ▶ Loan outcomes: Loan default frequency and charge-off rates
 - ▶ Bank costs: Net marginal expenses and fixed costs
 - ▶ Bank profitability: Dividend issuance, equity issuance
- ▶ We recalibrate bank fixed costs and the nonbank marginal cost change to match bank fixed costs and nonbank market share post-Dodd-Frank. [▶ Recalibrated Parameters](#)
- ▶ Model Properties: [▶ Transition](#) [▶ market shares](#)

[▶ Back Model Essentials](#)



PARAMETERS AND TARGETS

Parameter		Value	Target
Deposit Interest Rate (%)	r^D	0.0014	Avg Int Exp. Deposits (1984-2019)
Return on Securities	r_A	0.018	Return on Net Securities (1984-2019)
Convenience Yield	Υ_D	0.0166	$r_A - r^D$
Bank Discount Factor	β	0.9823	$(1 + r_A)^{-1}$
Default Frequency (Good Times)	θ_G	0.968	Mean Default Freq. (Non-Crisis)
Default Frequency (Bad Times)	θ_B	0.92	Mean Default Freq. (Financial Crisis)
Loan Loss Rate	$\bar{\lambda}$	0.31	Average Charge-off rate (1984-2019)
Standard Dev Loan Loss rate f	σ_λ	0.145	Std dev charge off rates f (1984-2019)
Standard Dev Loan Loss rate d	σ_λ	0.13	Std dev charge off rates d (1984-2019)

[▶ more parameters](#)
[▶ Back](#)

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PARAMETERS AND TARGETS (CONT.)

▶ BACK

Parameter		Value	Target
Size of deposit shocks d	$\{\epsilon_d^-, \epsilon_d^+\}$	$\{-0.0515, 0.0074\}$	Std. of shocks to deposits d
Size of deposit shocks f	$\{\epsilon_f^-, \epsilon_f^+\}$	$\{-0.44, 0.0383\} \times D_f$	Std. of shocks to deposits f
Probability of negative shock d	ρ_d^ϵ	0.125	$Pr(\epsilon_d < -\text{std})$
Probability of negative shock f	ρ_f^ϵ	0.08	$Pr(\epsilon_f < -\text{std})$
Number of Borrowers	N	5.05	Deposit to Output
Return on investing	R	0.13	Net Interest Margin
Price coefficient	α	37.0	Elasticity of Loan Demand
Lower bound demand shock	$\underline{\omega}$	0.0	Normalization
Upper bound demand shock	$\bar{\omega}$	0.28	Dividend Issuance d and f
Mean size of Dominant Bank d	D_d	0.15	Deposit Market Share Top 4
Mean size of Fringe Bank f	D_f	0.00039	Relative Size Fringe to Top 4
Linear Cost Loans d	C_d^1	0.007	Net Marginal Expenses Top 4
Quadratic Cost Loans d	C_d^2	0.00012	Elasticity Net Marginal Expenses Top 4
Fixed cost d	κ_d	0.00315	Fixed cost over loans Top 4
Mean Dist Cost Loans f	C_f^1	0.0068	Net Marginal Expenses Fringe
Quadratic Cost Loans f	C_f^2	$\frac{0.005}{D_f}$	Elasticity Net Marginal Expenses Fringe
Fixed cost f	κ_f	$0.0195 \times D_f$	Fixed cost over loans Fringe
External finance param. d	ψ_d^1	0.5	Avg. equity issuance to loan ratio Top 4
External finance param. f	ψ_f^1	3.15	Avg. equity issuance to loan ratio Fringe
Entry finance param. f	ψ_e	0.3	Fringe Bank Entry Rate
Entry Cost d	K_d	0.06	Q Top 4 Banks
Entry Cost f	K_f	$0.08 \times D_f$	Q Fringe Banks
Marginal Cost Nonbank	c_N	0.0182	Bank Loan Interest Rate
Nonbank Quality	ξ_N	0.225	Bank Loan to Total Loans Ratio



PARAMETERS THAT CHANGE

TABLE: Parameter Changes Post-Dodd-Frank

Parameter		Pre-Riegle-Neal	Post-Riegle-Neal	Post-Dodd-Frank
Regulatory Merger Cost	h	-	$0.0019 \times D_f$	$0.0019 \times D_f$
Nonbank Quality	ξ_N	0.225	0.475	0.475
Fixed cost d	κ_d	0.0032	0.0032	0.0053
Fixed cost f	κ_f	$0.0187 \times D_f$	$0.0187 \times D_f$	$0.0189 \times D_f$

► Back

Moment		Data		Model	
		Pre Riegle-Neal	Post Dodd-Frank	Pre Riegle-Neal	Post Dodd-Frank
Average Charge-off	$E_0'[(1 - \theta')\lambda']$	0.96%	0.94%	1.04%	1.04%
Std. of Deposit Shocks Top 4	$Std(\log(D_d + \epsilon_d))$	0.148	0.044	0.16	0.05
Std. of Deposit Shocks Fringe	$Std(\log(D_f + \epsilon_f))$	0.182	0.156	0.17	0.17
Freq. Neg. Deposit Shocks Top 4	$P(\epsilon_d < -\sigma_d)$	0.125	0.278	0.12	0.12
Freq. Neg. Deposit Shocks Fringe	$P(\epsilon_f < -\sigma_f)$	0.081	0.085	0.08	0.08
Elasticity of Loan Demand	$-\alpha r_D^{\frac{1}{\alpha}}(1 - s_B)$	-1.1	-1.1	-0.99	-1.14
Deposit Market Share Top 4	$\frac{D_d}{D_d + D_f}$	14.40%	44.76%	13.61%	41.66%
Bank Loans to Total Loans	$\frac{L_d + L_f}{L_d + L_f + L_I}$	44.54%	33.28%	44.44%	37.48%
Deposit to Output Ratio	$\frac{D_d + D_f}{\sum Output}$	39.01%	57.19%	55.68%	52.75%
Net Interest Margin	$E_0[\theta' r_D^{\frac{1}{\alpha}} - r_D]$	4.94%	4.35%	4.53%	4.63%
Net Expenses Top 4	$c'(L_D)$	1.16%	0.92%	0.73%	0.74%
Net Expenses Fringe	$c'(L_F)$	1.89%	1.45%	1.54%	1.63%
Elasticity Net Expenses Top 4	$\frac{d \log(C_d(L_d))}{d \log(L_d)}$	0.97%	1.05%	1.00%	1.00%
Elasticity Net Expenses Fringe	$\frac{d \log(C_f(L_f))}{d \log(L_f)}$	0.80%	0.87%	1.39%	1.41%
Fixed cost/loans Top 4	$\frac{F_d}{L_d}$	0.90%	0.60%	2.15%	2.40%
Fixed cost/loans Fringe	$\frac{F_f}{L_f}$	0.85%	0.56%	2.72%	2.47%
Equity Issuance to Assets Top 4	$\frac{1}{D_d}(\pi_d < 0)(-\pi_d)$	0.07%	0.01%	0.10%	0.00%
Equity Issuance to Assets Fringe	$\frac{1}{D_f}(\pi_f < 0)(-\pi_f)$	0.13%	0.04%	0.30%	0.29%
Bank Failure Rate Top 4	$\frac{F_d}{D_d}$	0.00%	0.00%	0.00%	0.00%
Bank Failure Rate Fringe	$\frac{F_f}{D_f}$	0.76%	0.29%	0.49%	0.29%
Relative Size Dominant/Fringe	$\frac{D_d}{D_f}$	415.80	998.31	384.62	1154.65
Dividends/Assets Top 4	$\frac{\partial D_d}{\partial V_d}$	0.36%	0.74%	0.70%	1.07%
Dividends/Assets Fringe	$\frac{\partial D_f}{\partial V_f}$	0.39%	0.66%	0.35%	0.42%
Tobin's Q Top 4	V_d/K_d	0.87	1.55	0.96	4.24
Tobin's Q Fringe	V_f/K_f	1.19	1.25	0.71	0.99



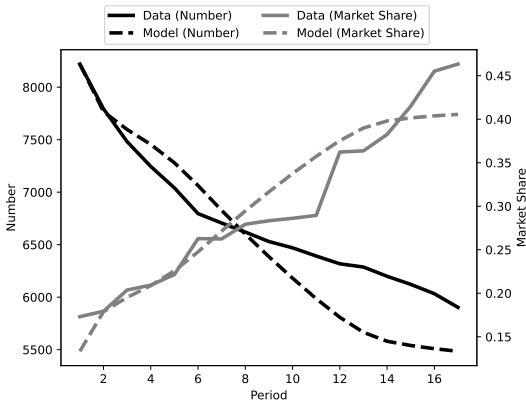
MOMENTS MODEL VS DATA (CONT.)

Moment		Data		Model	
		Pre Rieggle-Neal	Post Dodd-Frank	Pre Rieggle-Neal	Post Dodd-Frank
Loan Market Share Top 4	$\frac{L_d}{L_d + \gamma L_f}$	17.04%	41.46%	18.58%	31.32%
Bank Merger Rate	$\frac{(\gamma - \Gamma)}{\gamma}$	1.27%	2.89%	0.00%	0.17%
Loan Markup Top 4	$\frac{\theta r^L}{r_D + c'(L_d)} - 1$	56.17	221.46	434.53	445.36
Loan Markup Fringe	$\frac{\theta r^L}{r_D + c'(L_f)} - 1$	47.82	152.57	178.55	169.52
Interest Rate	r^L	6.69%	3.18%	4.83	4.94
Share of Deposits Lent Top 4	$\frac{L_d}{D_d}$	78.01%	63.23%	99.98%	48.97%
Share of Deposits Lent Fringe	$\frac{L_f}{D_f}$	67.60%	73.26%	70.13%	77.53%

► Back



MODEL PROPERTIES: TRANSITION PATH



- Model captures the decline in the number of banks and the growth of dominant bank market share of bank lending (as in Fig 1). [return](#)

[return model essentials](#)



PRICE AND MARKET SHARE CHANGES

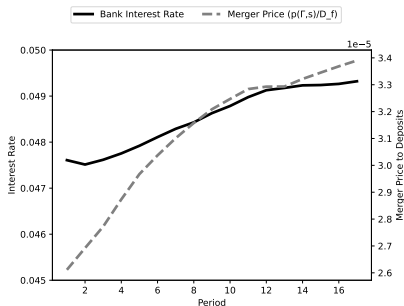


FIGURE: Loan Rate and Merger Price

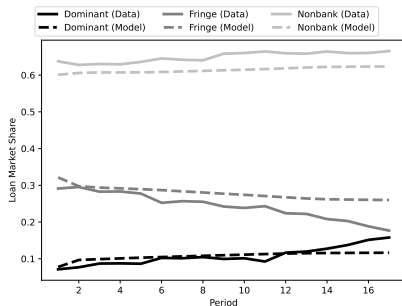


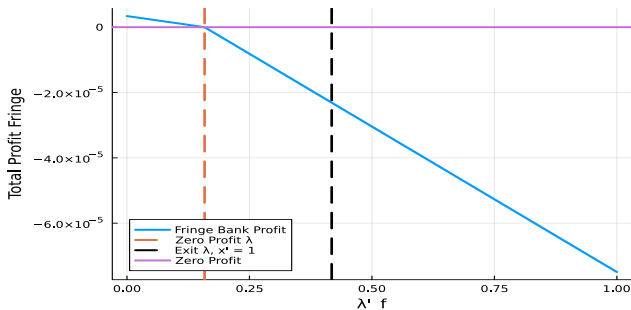
FIGURE: Loan Market Shares

- ▶ Bank mergers increase the price of loans (+6bp) and the value of fringe banks
 - ▶ [return](#)
 - ▶ [return model essentials](#)



CHARGEOFFS AND FINANCIAL STABILITY

We plot a fringe bank's realized profit (after costly equity issuance) as a function of the realization of λ in the worst realization of ϵ_f and θ' .

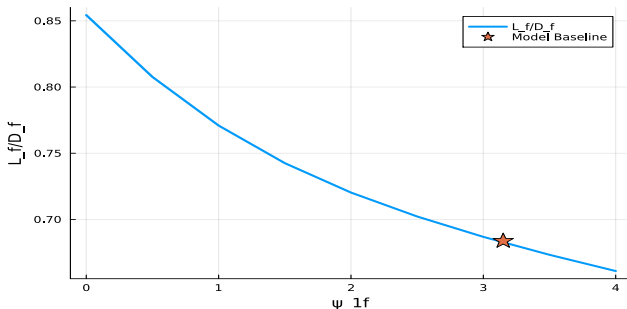


- ▶ The kink in the line at $\lambda_f = 0.10$ is where the fringe bank now has to issue costly equity.
- ▶ The high exit threshold means the fringe bank prefers to issue costly equity as opposed to exiting. [▶ return](#)



COSTLY EXTERNAL FINANCE AND LENDING

We plot the fringe bank's lending policy, as a function of how costly it is to issue equity, ψ_f .



- ▶ The fringe bank decreases lending as the cost of equity issuance rises because lending is risky.
- ▶ The larger MPL in the Pre-Riegle-Neal calibration for the dominant bank may be due to the dominant bank's lower cost of equity issuance.

[return](#)



GRANULAR REGRESSIONS - TOTAL CREDIT

- ▶ We run granularity regressions in the model in pre-Riegle-Neal and post-Dodd-Frank environments.

TABLE: Model-based granular regressions (Total Credit)

Coefficient	Pre Riegle-Neal	Post Riegle-Neal Short-run	Post Riegle-Neal Long-run
Intercept	-0.0	-0.001	-0.0
Shock Coefficient	0.28	0.023	0.035
R^2	0.484	0.017	0.113
N	9000.0	15000.0	9000.0

- ▶ Model explains 32% (Pre) and 10% (Post) of total credit (> data pre-reform) [▶ data](#)
- ▶ While market share of dominant banks increase, the decline in R^2 is due to the decline in MPL and the participation of nonbanks [▶ return](#)

COUNTERFACTUAL

[▶ RETURN](#)

Effect of Mergers **Comparing columns 1 and 2**

- ▶ Allowing mergers leads to a more concentrated banking sector
- ▶ Decreased competition leads to an increase in interest rates and a jump in bank values

Effect of Nonbank Quality Improvement **Comparing columns 2 and 3**

- ▶ Improved nonbank competition leads to less bank lending
- ▶ Competition with improved nonbanks leads to a significant decline in loan interest rates (which also results in higher failure rates)
- ▶ Dominant value relatively unchanged, while fringe value declines

Effect of DF Regulation **Comparing columns 3 and 4**

- ▶ Bank value and bank lending decline leading to a small increase in the failure rate

Effect of everything **Comparing columns 2 and 4**

- ▶ Higher nonbank quality together with the increase in regulatory pressure on banks leads to a decline in bank loan concentration and a shift towards nonbank lending.
- ▶ Interest rates and markups for large banks decline slightly
- ▶ The value of the dominant and fringe banks decline.
- ▶ Financial stability declines as failure rate of small banks increases

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COUNTERFACTUAL: RIEGLE-NEAL AND DODD-FRANK

	Pre RN	Post RN ($\downarrow H$)	Post RN ($\downarrow H \uparrow \xi_k$)	Post DF ($\downarrow H \uparrow \xi_k \uparrow \kappa_b$)
Initial Γ	2500.00	1667.45	1667.45	1667.45
Initial D_d	0.15	0.44	0.44	0.44
Deposit Share Top 4	13.61%	41.15%	41.54%	41.66%
Loan Share Top 4	18.58%	31.65%	31.39%	31.32%
Bank Lending Share	44.44%	42.88%	37.51%	37.48%
Bank Interest Rate	4.83%	5.01%	4.94%	4.94%
Nonbank Interest Rate	4.84%	4.84%	4.84%	4.84%
Share Lent Top 4	99.98%	54.26%	49.14%	48.97%
Share Lent Fringe	70.13%	82.59%	77.18%	77.53%
Fringe Failure Rate	0.49%	0.11%	0.20%	0.29%
Merger Rate	0.00%	0.18%	0.17%	0.17%
Tobin's Q Top 4	0.96	5.74	5.67	4.24
Tobin's Q Fringe	0.71	1.30	1.09	0.99
Household Value	3.46	5.11	5.26	4.76
Borrower Value	849.83	848.65	847.30	847.28

MONETARY POLICY

- ▶ Consider a temporary, unanticipated monetary policy shock that increases r^A by 25bp and r^D by 10bp (similar to Dreschler et al. (2017)).
- ▶ There is 50% chance of reverting to baseline monetary policy every period.

▶ return

Moment	Pre-Riegle-Neal	Post-Riegle-Neal	Post-Dodd-Frank
Total Loans by Dominant Bank	-0.01%	-0.94%	-0.93%
Total Loans by Fringe Banks	-0.26%	-0.01%	+0.05%
Total Bank Loans	-0.21%	-0.25%	-0.26%
Total Loans	-0.44%	-0.44%	-0.45%
Bank Loans to Total Loans	+0.24%	+0.20%	+0.20%
Share of Deposits Lent Dominant	-0.01%	-1.31%	-0.93%
Share of Deposits Lent Fringe	-0.34%	-0.01%	-0.06%
Bank Interest Rate	+4.0bp	+4.3bp	+4.3bp
Number of Fringe Banks	+0.06%	-0.03%	+0.14%
Size Dominant Bank	-0.00%	+0.39%	-0.00%
Entry Rate	-1.7bp	-0.0bp	-0.0bp
Failure Rate (Fringe)	-1.8bp	-1.4bp	-1.3bp
Tobin's Q (Fringe)	+0.32%	+0.39%	+0.24%
Tobin's Q (Dominant)	+0.02%	+0.08%	-0.24%

- ▶ Banks with smaller MPL reduce lending the most - fringe banks Pre-RN and



DEMAND FOR LOANS

- ▶ A mass $\mathcal{E}$ of ex-ante identical borrowers demand one-period loans to fund a risky project.
- ▶ Every period, given r_B^L, r_N^L , and shock ω , borrowers decide whether to invest ($\iota = 1$) or not ($\iota = 0$).

$$\max_{\{\iota\}} (1 - \iota) \cdot \omega + \iota \cdot E_{\delta}[\Pi_E(\theta, r_B^L, r_N^L, \delta)] \quad (21)$$

- ▶ Conditional on choosing $\iota = 1$, entrepreneurs observe $\delta = \{\delta_B, \delta_N\}$ and $\xi = \{\xi_B, \xi_N\}$ and then choose which type of lender $k \in \{B, N\}$ to borrow from to solve:

$$\Pi_E(\theta, r_B^L, r_N^L, \delta) = \max_{k \in \{B, N\}} \alpha E_{\theta'|\theta}[\pi(r_k^L, \theta)] + \xi_k + \delta_k \quad (22)$$

where

$$\pi_E(r_k^L, \theta') = \begin{cases} \max\{0, R - r_k^L\} & \text{with prob } \theta' \\ \max\{0, -(\lambda' + r_k^L)\} & \text{with prob } 1 - \theta' \end{cases}$$



DEMAND FOR LOANS (CONT.)

- The solution implies market shares for lender of type k :

$$\psi_k(\theta, r_B^L, r_N^L) = \frac{\exp(\alpha E_{\theta'|\theta}[\pi_E(r_k, \theta')] + \xi_k)}{\sum_{\hat{k} \in \{\mathcal{B}, \mathcal{N}\}} \exp(\alpha E_{\theta'|\theta}[\pi_E(r_{\hat{k}}, \theta')] + \xi_{\hat{k}})}.$$

- Total demand for credit is

$$L(\theta, r_B^L, r_N^L) = \mathcal{E} \int \iota(\theta, r_B^L, r_N^L, \omega) d\Omega(\omega)$$

- Bank loan demand

$$L_B(\theta, r_B^L, r_N^L) = \psi_B(\theta, r_B^L, r_N^L) L(\theta, r_B^L, r_N^L)$$

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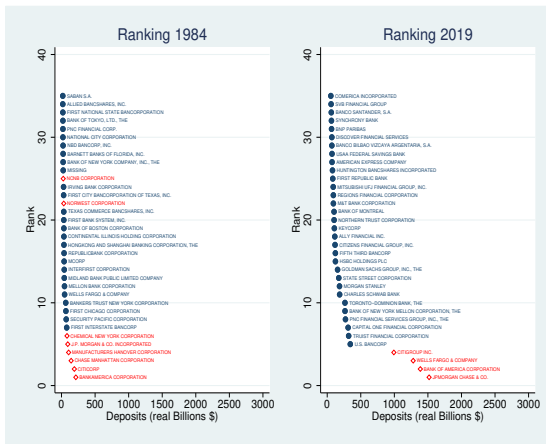


REGULATORY CONSTRAINT

$$H_{pre-RN}(\Gamma, s) = \infty \quad (23)$$

$$H_{post-RN}(\Gamma, s) = \begin{cases} \infty & \text{if } \frac{D'_d}{D_d + \gamma D_f} > 0.40 \\ h(\gamma - \Gamma)^2 & \text{if } HHI > 1800 \text{ or } \{HHI < 1800 \text{ and } \Delta_{HHI} > 10 \\ 0 & \text{otherwise} \end{cases} \quad (24)$$

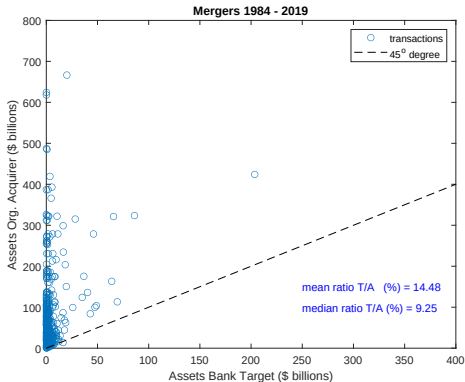
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- ▶ The Top 4 have grown even relative to other banks at the top of the deposit distribution



ACQUIRERS VS. TARGETS



- ▶ For the median merger, the acquirer is 9 times larger than the target.

▶ Pre & Post

▶ Distribution

- ▶ Acquirers are larger than targets whether measured by assets or market capitalization.

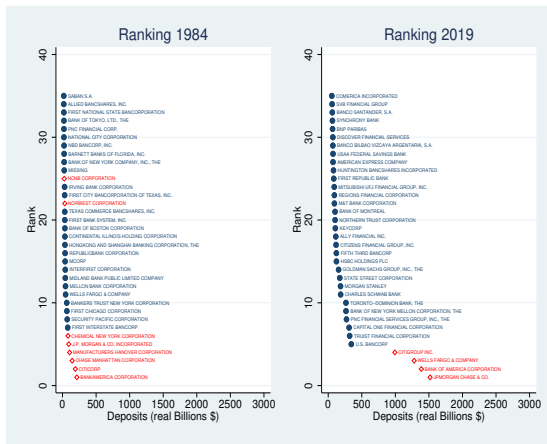
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GRANULAR BANKS AND AGGREGATE BANK LENDING

	Top 4 Banks - Dep. Var. $\Delta \log(L_t)$			
	Full Period 1984-2019	Pre-reform 1984-1993	Transition 1994-2007	Post-GFC 2011-2019
Γ_t^x	0.512*** (0.141)	-0.742 (0.515)	0.562*** (0.152)	-0.380 (0.351)
Γ_{t-1}^x	0.201 (0.141)	0.232 (0.509)	0.0496 (0.155)	0.0924 (0.352)
Γ_{t-2}^x	-0.0139 (0.142)	-0.749 (0.517)	-0.0267 (0.157)	-0.442 (0.353)
(intercept)	0.00834*** (0.00114)	0.00165 (0.00251)	0.0131*** (0.00143)	0.00478** (0.00233)
N	144	40	56	36
R^2	0.100	0.104	0.212	0.084

- R^2 of 0.10 suggests that idiosyncratic shocks to the largest 4 banks explain 10% of the variation in aggregate bank lending.

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[▶ back model](#)



- The Top 4 have grown even relative to other banks at the top of the deposit distribution ► Back

NONBANK COMPETITION

- We compare the effect of an increase in nonbank competition pre-RN and post-RN.

Moment	Pre-Riegle-Neal	Post-Riegle-Neal Short Run	Post-Riegle-Neal Long Run
Total Loans by Dominant Bank	-0.12%	-8.73%	-7.51%
Total Loans by Fringe Banks	-10.45%	-8.70%	-10.45%
Bank Loans to Total Loans	-12.23%	-12.44%	-13.18%
Share of Deposits Lent Dominant	-0.12%	-9.32%	-5.86%
Share of Deposits Lent Fringe	-7.23%	-6.60%	-3.39%
Bank Interest Rate	-7.3bp	-7.0bp	-5.7bp
Number of Fringe Banks	-4.84%	-2.31%	-7.44%
Size Dominant Bank	-0.00%	-1.31%	-1.58%
Entry Rate	+7.9bp	-4.9bp	-5.2bp
Failure Rate (Fringe)	+55.5bp	+13.5bp	+6.8bp
Tobin's Q (Fringe)	-23.49%	-20.09%	-9.49%
Tobin's Q (Dominant)	-3.59%	-1.43%	-0.97%

- Improved nonbank competition reduces financial stability. The effect of improved competition is stronger pre-Riegle-Neal, when fringe banks cannot be acquired.
- Prospect of being acquired leads fringe banks to take on more risk as measured by share of deposits lent. [► Back](#)

NORMATIVE EFFECTS OF RIEGLE-NEAL

- ▶ We ask the question “What would households and entrepreneurs be willing to pay (or be paid) to adopt Riegler-Neal?”

		Post-RN short-run	Post RN long-run
Household	α_H^T	29.19	39.88
Entrepreneur	α_E^T	-0.07	-0.11
Average	$\bar{\alpha}^T$	0.50	0.67

Note: $\{\alpha_H, \alpha_E\}$ reported in consumption equivalents.

The baseline economy is the long-run pre-RN [▶ details](#)

- ▶ Positive effects of RN (large for households). The increase in bank loan interest rates reduces entrepreneurs welfare but this is more than compensated by the increase in allocative efficiency (and markups), the reduction in the number of fringe banks (reduce fixed operating costs), and the reduction in failure rates.
- ▶ *Alternative merger policies*: changes in deposit cap and merger restrictions h . Negative welfare effects in the short run but positive in the long-run [▶ results](#)



NORMATIVE EFFECTS OF RIEGLE-NEAL

- ▶ We ask the question “What would households and entrepreneurs be willing to pay (or be paid) to adopt Riegle-Neal?”
- ▶ Let $C_{a,t}^{pre}$ denote type $a \in \{H, E\}$ equilibrium consumption in the pre Riegle-Neal economy. Let $C_{a,t}^{post}$ consumption post-reform
- ▶ The consumption equivalent α_a^T for a representative agent a is defined as the constant percentage change increase (or reduction) in consumption in the pre-reform that gives the agent the same expected utility

$$E_0 \left[\sum_{t=0}^T \beta_j^t C_{a,t}^{post} \right] = E_0 \left[\sum_{t=0}^{\infty} \beta_a^t (1 + \alpha_a^T) C_{a,t}^{pre} \right].$$

where we set $T \in \{16, \infty\}$ for short-run and long-run effects, respectively

- ▶ We let ex-ante welfare pre-RN be (post RN is defined similarly)

$$W^{pre} = \frac{\mathcal{H}}{1 + \mathcal{H}} E_0 \left[\sum_{t=0}^{\infty} \beta_H^t C_{H,t}^{pre} \right] + \frac{1}{1 + \mathcal{H}} E_0 \left[\sum_{t=0}^{\infty} \beta_E^t C_{E,t}^{pre} \right]$$

and average effect is $W^{post} = (1 + \bar{\alpha})W^{pre}$.

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ALTERNATIVE MERGER POLICIES

- We ask the question “What would households and entrepreneurs be willing to pay (or be paid) to move away from Riegle-Neal?”

Moment	Higher Cap (50%)	Lower Cap (30%)	Higher Cost ($\uparrow h$)	Lower Cost ($\downarrow h$)
Household C.E. α_H^T (%)	4.75	-3.02	-2.59	1.73
Entrepreneur C.E. α_E^T (%)	-0.03	0.02	0.01	-0.01
Average $\bar{\alpha}^T$ (%)	0.10	-0.06	-0.06	0.04

Note: $\{\alpha_H, \alpha_E\}$ reported in consumption equivalents. Baseline is Long-Run Equilibrium post Riegle-Neal. Short Run $T = 16$.

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